



Institute of Infrastructure Technology Research and Management

End Semester
(Start time: 10.00 a.m. and End time: 01:00 p.m.)
B. Tech. (All Branches)
Semester - VI

Date: 23/04/2025

Course Name: Infrastructure Planning and Management

Course Code : HS 225002

Max. Marks: 60

Instructions:

1. Course instructor may penalize the student by deducting the marks, if answer is adjudged to be copied during assessment. This act shall be recorded as UFM (unfair means).
 2. Assume suitable data wherever essential and mention it clearly.
 3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.
-

Section A

Attempt any three questions. Each question carries 10 marks. (10X 3 = 30 Marks)

- Q.1) Analyze the socio-environmental and legal issues faced by the Lavasa hill station township project as outlined in the case study. Discuss the violations of environmental laws, land acquisition controversies, and their impact on local communities and ecology.
- Q.2) Discuss the PPP model and its impact of infrastructure project. Briefly discuss the key lesson from the Delhi and Mumbai airport privatization bidding process that could improve future infrastructure project evaluations.
- Q.3) How can Bharti Airtel can compete with Reliance Jio's low-cost data strategy, while addressing the challenges of declining voice revenue and increasing debt in the Indian telecom sector after 2017?
- Q.4) Discuss the risk management process for infrastructure projects. Identify key risks in infrastructure project and evaluate their corresponding mitigation strategies
- Q5) Critically analyze the economic, environmental, and social impacts of the Golden Quadrilateral project. How have the engineering innovations and integration with national infrastructure initiatives like Bharatmala and the Delhi-Mumbai Expressway contributed to its success? Discuss with examples



Institute of Infrastructure Technology Research and Management

Section B

Attempt all the questions. Each question carries 10 marks. (15 X 2 = 30 Marks)

Q.1) L & T Infrastructure Company is planning to undertake an infrastructure project with an initial investment of ₹25,00,000. Consider the following cash flow from the project and suggest whether Adani Infrastructure should invest at 12%, 15%, and 18% discount rates. Why and why not?

Year	2023	2024	2025	2026
Cash Flow	₹5,00,000	₹8,00,000	₹12,00,000	₹15,00,000

Instructions:

- Calculate the Net Present Value (NPV) for each discount rate.
- Provide a recommendation based on whether the NPV is positive or negative.
- Explain your reasoning for each discount rate.

Q.2) Reliance Infrastructure Company is planning to build a toll highway with an initial investment of ₹1,200,000. The decision to invest depends on the state of the economy and the resulting traffic volume. The probabilities for the state of the economy are as follows: Good (0.30), OK (0.40), and Bad (0.30). The traffic volumes (High, Medium, Low) are conditional on the state of the economy based on expert analysis. The payoffs for different traffic volumes are ₹1,600,000 (High), ₹1,300,000 (Medium), and ₹600,000 (Low). The expert-provided conditional probabilities are:

Economy vs Traffic	Good	OK	Bad
High	0.7	0.5	0.2
Medium	0.2	0.3	0.4
Low	0.1	0.2	0.4

Will Reliance Infrastructure Company invest in the toll highway? Why or why not? Calculate the Expected Value (EV) to support your decision.